

Modeling Cell-Type-Specific MMP Dynamics and ECM Remodeling in the Tumor Microenvironment

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BACKGROUND RESEARCH

Metastasis is the leading cause of cancer mortality. It is a multi-step process that includes invasion of adjacent tissue by tumor cells, entry into the vasculature, and exit of carcinoma cells from the circulation and colonization at the distal sites (1). The invasion of cancer cells is greatly affected by the tumor microenvironment, consisting of stromal cells, immune cells, and the extracellular matrix (ECM). The ECM is composed of proteins like collagen and other glycoproteins. Matrix metalloproteinases (MMPs) belong to a zinc-dependent family of endopeptidases and play a big role in the ECM remodeling of the tumor microenvironment during the initial invasion by tumor cells (2). They degrade the ECM and facilitate invasion and angiogenesis. They also increase the availability of growth factors in the microenvironment by liberating them from the ECM (2).

The action of MMPs is tightly regulated. TIMPs inhibit MMPs by forming tight 1:1 stoichiometric complexes with the active site of MMPs, thereby preventing them from degrading ECM components. This regulatory mechanism is crucial for normal tissue remodeling and for preventing excessive matrix degradation (3).

According to a review article we found (4), based on their effect on tumor progression, MMPs can be classified as having pro-metastatic, anti-metastatic, and dual roles. MMP2 and MMP8 are pro- and anti-metastatic, respectively. MMP2 is released by tumor cells and cleaves type IV collagen and is associated with poor prognosis (5). On the other hand, MMP8 is released by neutrophil cells and has been shown to have a protective role (6). It has been found that MMP8 functions as a metastasis suppressor by increasing cell-ECM adhesion (7). Apart from this, MMP8 can degrade decorin, which acts as a TGF- β reservoir, hence affecting EMT. (8)

Apart from this, ECM remodeling by MMPs can affect the effectiveness of tumor cells. As found in a research by H  l  ne S et al, Matrix architecture defines the preferential localization and migration of T cells into the tumor stroma (9)

PROBLEM STATEMENT

After extensive research, we have formulated a problem statement to work on: Develop an agent-based model integrating pro-tumorigenic and anti-tumorigenic MMPs and TIMP secretions by tumor and immune cells, and check the effect of their interactions on tumor invasion.

Objectives:

1. Model tumor and immune cell interaction in the presence of MMP and TIMP by varying their diffusion rates.
2. Check the effect of MMP8 on these interactions

METHODOLOGY

Modeling approach:

For our project, we chose agent-based modeling to capture the MMP secretion by individual cells and the difference in the tumor migration due to varying cell-ECM and cell-cell adhesion strengths. We used CompuCell3D to achieve this. CompuCell3D (CC3D) is an agent-based simulation platform designed for modeling complex biological systems, particularly in the fields of cell biology, developmental biology, and cancer research. It is primarily built upon the Cellular Potts Model (CPM), also known as the Glazier-Graner-Hogeweg (GGH) model (11). CC3D integrates the CPM/GGH framework with partial differential equation (PDE) solvers to simulate chemical fields and other dynamic processes. The framework enables the spatial representation of both cellular components and non-cellular components

CC3D updates the model based on its total energy, which is calculated by the following equation:

$$H = \sum [J(\tau(\sigma(i)), \tau(\sigma'(j)))(1 - \delta(\sigma(i), \tau(\sigma'(j))))] + \sum [\lambda(\sigma) (v(\sigma) - v_T(\sigma))^2] + H_{chemo} + H_{AM}$$

During each Monte Carlo Step (MCS), the simulation attempts to copy the index (identity) of one pixel to a neighboring pixel. When such a copy attempt occurs, the system calculates the change in effective energy resulting from this potential update. If the change in energy (ΔH) is negative, meaning the new configuration is more energetically favorable, the copy is automatically accepted, and the update will be reflected in the next MCS. If ΔH is positive, indicating a less favorable configuration, the copy may still be accepted, but only with a certain probability defined by the Boltzmann acceptance function:

$$P = e^{-\Delta H/T}$$

Here, T represents the system's effective temperature, which controls the likelihood of accepting energetically unfavorable changes. This probabilistic mechanism allows the simulation to occasionally explore higher-energy states, preventing it from getting trapped in local energy minima and enabling more realistic dynamic behavior.

We divided the modeling process as follows:

1. Model ECM degradation by cancer cells secreting MMP
2. Add immune cells to the model and observe the interaction between tumor cells and immune cells in the context of ECM degradation by MMP.
3. Incorporate MMP-8 secretion by immune cells and observe the invasion of tumor cells due to the interplay of the pro- and anti-metastatic MMPs.

Due to time constraints, we could only model the first two cases. To model the ECM degradation by cancer cells secreting MMP, we tried reproducing the results of a paper by Kumar S. et al., in 2016 (10). We proceeded with the second step by combining two different models by Prasanna et al., in 2024 and Emma L. et al. in 2024 (11)(12). Prasanna's model looks at the change in collective cell invasion with a change in tumor heterogeneity. To his model, we added immune cells, whose parameters were taken from Emma L.'s model. Our final model had the following components:

1. Tumor cells- Our model consists of two kinds of cancer cells: Cancer1 and Cancer2 cells. Cancer1 cells have high cell-cell adhesion and low cell-ECM adhesion, hence mimicking epithelial cells. Cancer2 cells have low cell-cell adhesion and high cell-ECM adhesion, hence mimicking mesenchymal cells.

A MitosisSteppable is used for the division of tumor cells into daughter cells.

2. Immune cells- Immune cells are formed at the periphery of the model. They kill off cancer cells according to a killing probability.
3. ECM- The ECM is composed of laminin and collagen. Only laminin is degraded to form l_ysed, which is an intermediate lysed form of laminin.

And the following chemicals:

1. MMP and TIMP- The cancer cells release MMP and TIMP on contact with Laminin. MMP degrades the ECM, and TIMP inhibits the action of MMP. The degradation of the ECM occurs if the ratio of MMP concentration to TIMP concentration is more than 2.
2. Growth Factor (GF)- The ECM secretes the growth factor. Laminin cells secrete it on contact with a cell. The target volume of cancer cells increases as a function of the concentration of the growth factor.
3. Chemoattractant- Chemoattractant is secreted by the cancer cells, and it attracts immune cells in the direction of its higher concentration.

The parameters used for the model are included in the supplementary.

The quantification of the invasion by tumor cells is done by the following parameters:

1. Cell Count: This is the total number of cells of a given cell type
2. Cell Invasion: This is calculated as the total volume of all the cells of a particular type. The rationale behind this is that when tumor cells get more space, they can grow more efficiently and hence are more invasive, therefore making their volume a good proxy of invasion.

RESULTS

We investigated how varying the diffusion constant of MMPs (from 0.005 to 0.05) and tumor cell population sizes impacts tumor invasion and ECM remodeling in the tumor microenvironment using our CompuCell3D model.

Legend to be followed:

Cell Type Id	Color	Name	Show in
0	Black	Medium	☐
1	Blue	CoI	☒
2	Green	Laminin	☒
3	Red	Cancer1	☒
4	Yellow	Cancer2	☒
5	Brown	C_Lysed	☒
6	Light Blue	L_Lysed	☒
7	Light Green	NCoI	☒
8	Purple	Immune...	☒
9	Orange	Apoptoti...	☒

A. Effect of MMP Diffusion Constant on Tumor Invasion and ECM Remodeling

Four different diffusion constants were tested:

- 0.005: The tumor invasion is irregular and poorly controlled. MMP diffusion is highly localized, leading to uneven ECM degradation. The inhibitor also spreads weakly and fails to regulate MMP activity effectively. This results in chaotic and spatially disorganized invasion.
- 0.01: A compact and directed invasion is observed. MMP forms a strong central gradient, and the inhibitor is sufficiently diffused to counteract MMP, leading to regulated ECM degradation.
- 0.02: The invasion becomes broader but remains spatially coherent. MMP and inhibitor fields are slightly more dispersed yet still centered, allowing enhanced but stable invasion.
- 0.05: Tumor invasion becomes excessive and erratic. MMP diffuses too widely, forming multiple degradation sites. The inhibitor becomes too diluted to be effective, causing loss of spatial control.

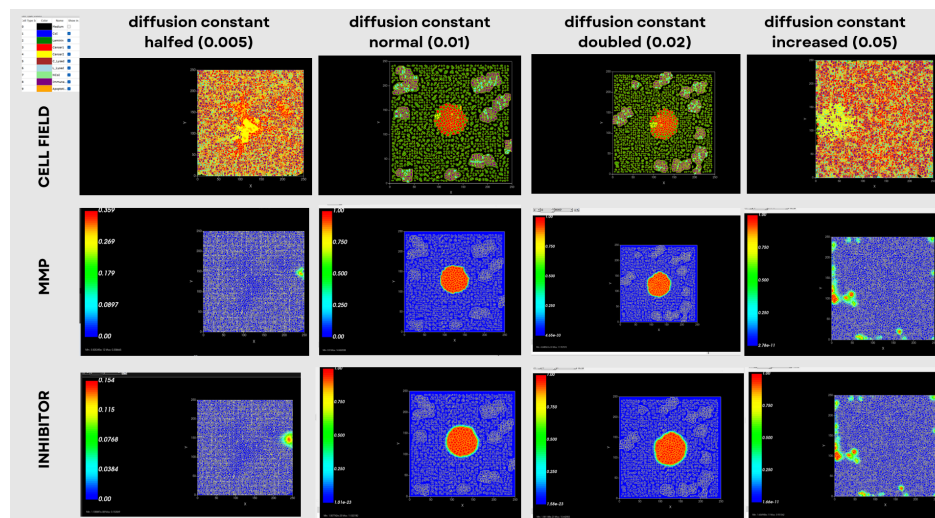


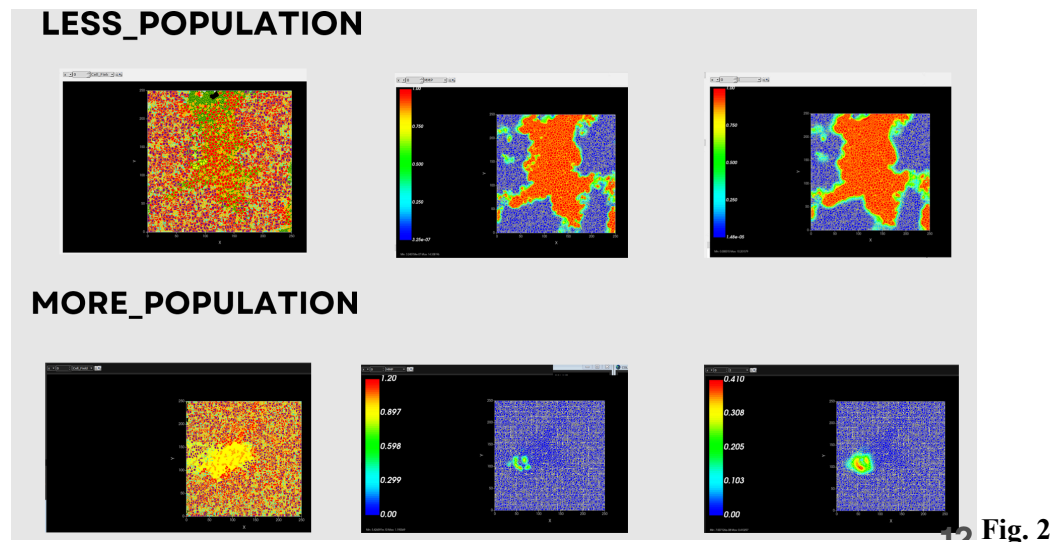
Fig. 1

B. Effect of Tumor Cell Population on Invasion and ECM Remodeling

The impact of tumor cell density is analyzed under three conditions:

- Less population: Sparse tumor invasion with broad MMP spread and uncontrolled ECM degradation. The inhibitor fails to spatially regulate MMP, resulting in poorly confined invasion.
- Intermediate population: Tumor forms a compact mass with a focused MMP gradient near its boundary. The inhibitor's distribution supports spatial regulation, leading to moderately strong and controlled invasion.

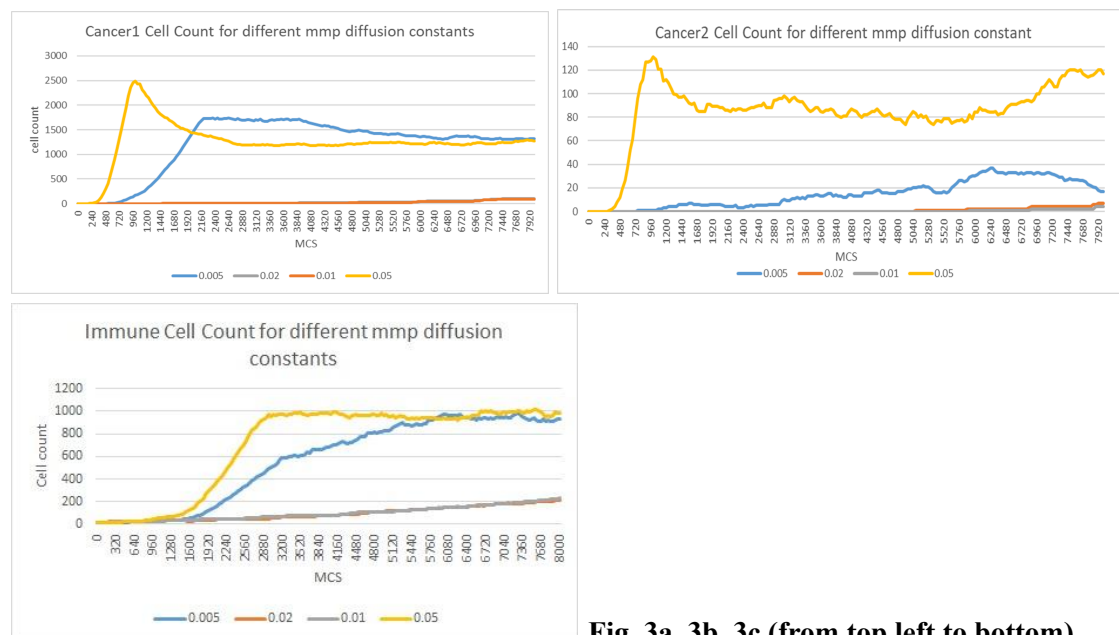
c. More population: Tumor mass is dense and cohesive. MMP concentrates near the core, and the inhibitor effectively restrains its spread. Invasion is localized and boundaries are well defined.



C. Cell Count Analysis for Different MMP Diffusion Constants

Line graphs (Fig 3a,3b,3c) show the trends in tumor and immune cell counts over time for varying MMP diffusion constants:

- a. 0.01 and 0.02: Cancer cell counts increase gradually and remain regulated.
- b. 0.05: Rapid growth is followed by a decline, likely due to increased immune activity at the tumor periphery.
- c. 0.005: Unexpectedly high cancer cell counts are observed despite limited MMP diffusion. This could be due to reduced immune infiltration and the localized nature of ECM degradation, enabling unchecked tumor expansion.



D. Cancer Invasion Volume Across MMP Diffusion Rates

Bar graphs (fig 4a, 4b) compare the invasion volumes of Cancer1 and Cancer2 cells under different MMP diffusion constants:

- a . Invasion is highest at 0.005, despite expectations, and lower at 0.01 and 0.02, where regulation is strongest.
- b. At 0.05, invasion increases again but decreases eventually, potentially due to the presence of more immune cells.

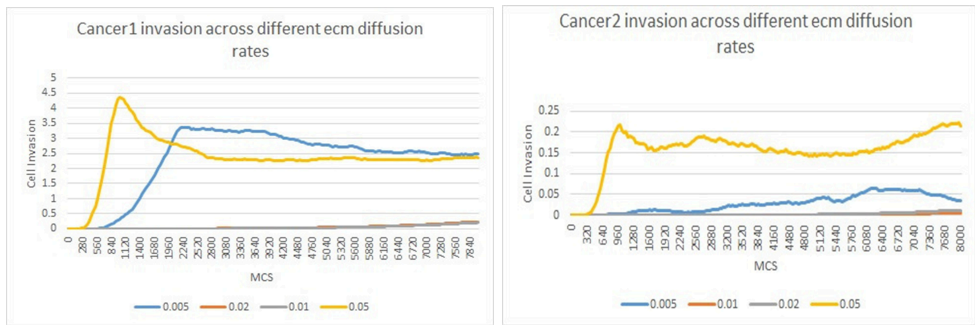


Fig. 4a, 4b

E. Correlation Insights

This section interprets underlying relationships between model components:

- a. Mutation effects strongly correlate with Cancer2 (Cancer B) growth in both count and area.
- b. Immune cell numbers rise with time and mutation level, but higher immune presence corresponds with lower cancer burden.
- c. Cancer1 and Cancer2 behaviors are tightly linked, showing coordinated expansion.
- d. High immune-to-cancer cell ratios and increased cell area correlate with reduced tumor load, indicating immune suppression and spatial crowding.

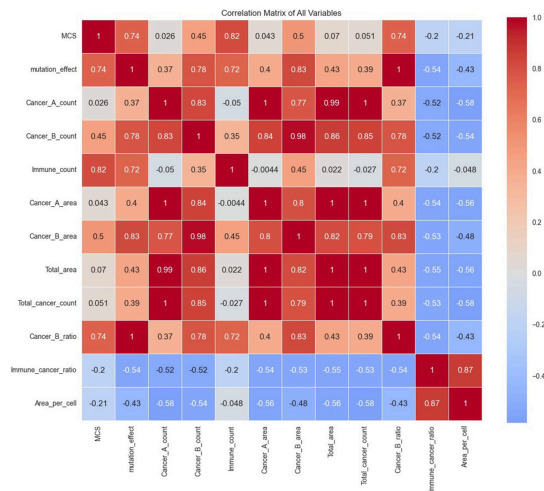


Fig. 5

CONCLUSION

Our study demonstrates that tumor invasion and ECM remodeling are significantly influenced by both the diffusion properties of MMPs and the size of the tumor cell population. Using an agent-based CompuCell3D model, we showed that intermediate diffusion of MMPs leads to more spatially regulated and directed invasion, whereas very low or very high diffusion results in disorganized or excessive ECM

degradation. Similarly, we found that tumor population size modulates the spatial patterns of MMP activity and invasion, with denser populations producing more cohesive and controlled invasion fronts.

Unexpected results at the lowest MMP diffusion constant suggest that spatial confinement and immune suppression dynamics may produce non-intuitive behaviors, emphasizing the importance of accounting for cell–field and cell–cell interactions in a spatial context. Furthermore, our analysis of Cancer1 and Cancer2 populations highlights how epithelial–mesenchymal heterogeneity affects invasive potential, with epithelial-like Cancer1 cells showing higher invasion; however, their invasion shows a greater decrease than mesenchymal-like Cancer2 cells throughout the simulation for MMP diffusion constant 0.05.

These results collectively underscore the importance of MMP–TIMP balance and tumor heterogeneity in shaping invasion dynamics. The model provides a platform to further investigate the roles of additional MMP types, such as MMP8, and to explore therapeutic strategies targeting ECM remodeling and immune modulation in the tumor microenvironment.

DISCUSSION

This study shows that tumor invasion is strongly influenced by MMP diffusion and tumor cell population size. Intermediate MMP diffusion led to well-regulated, directional invasion, while very low or high diffusion caused disorganized growth. Unexpectedly, low diffusion (0.005) resulted in higher cancer cell counts, possibly due to limited immune response and localized ECM degradation. Invasion by Cancer2 cells varies less than Cancer1 throughout the simulation, reflecting their mesenchymal traits. The model simplifies cell behavior and signaling, omitting processes like EMT plasticity and immune exhaustion. Chemical regulation is also oversimplified, and no experimental validation was performed. Future work could add MMP8 to explore anti-metastatic effects from immune cells. Including stromal cells and more detailed signaling could enhance biological realism. Parameter uncertainty and stochastic variability should also be addressed. Despite limitations, our model provides insights into how ECM remodeling shapes tumor progression.

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